Student Performance Analysis

Project Report — Data Analysis Practical

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Project	Student Performance Analysis
Subject	Data Fundamentals
Tools Used	Python (Pandas, Matplotlib) | PostgreSQL | Microsoft Excel
Dataset	students.csv — 5 students, 7 attributes

1. Problem Statement

Educational institutions collect large amounts of student data but often lack the tools to turn that data into actionable insights. Without proper analysis, it is difficult for academic committees to identify struggling students early, compare department performance, or understand how attendance affects academic outcomes.

This project analyses the academic performance of five students from the Computer Science (CS) and Information Technology (IT) departments. The dataset contains marks in Mathematics, Science, and Programming along with attendance records. The goal is to derive meaningful insights that can support data-driven academic decisions.

Key Questions Answered

- Who is the top-performing student?
- Which department performs better academically?
- Which students are at risk due to low marks?
- Is there a relationship between attendance and performance?
- What is the overall class average?

2. Dataset

The dataset used is students.csv, containing records of 5 students across two departments. It includes 7 attributes: student ID, name, department, marks in three subjects, and attendance percentage. An additional average_marks column was derived during analysis as (Math + Science + Programming) / 3.

ID	Name	Dept	Math	Science	Prog	Attendance	Avg Marks
4	Pooja	IT	92	91	95	96%	92.67
1	Amit	CS	78	85	90	92%	84.33
2	Riya	CS	88	79	85	90%	84.00
3	Karan	IT	65	70	72	80%	69.00
5	Rahul	CS	55	60	58	70%	57.67

Table 1 — Student dataset sorted by average marks (red cells = below 60)

3. Solution and Features

The analysis was carried out using three tools — SQL for querying, Microsoft Excel for spreadsheet analysis, and Python (Pandas + Matplotlib) for programmatic analysis and visualization. Each tool was used to answer the same business questions from a different perspective.

3.1 SQL Analysis

Six SQL queries were written to extract insights directly from the dataset:

Task	SQL Query	Result
1. Display Records	`SELECT * FROM students;`	All 5 rows returned
2. Avg per Student	`ROUND((math+science+prog)/3.0, 2) AS avg`	Per-student averages
3. Top Student	`ORDER BY avg DESC  LIMIT 1;`	Pooja — 92.67
4. Dept Average	`GROUP BY department + AVG()`	CS: 75.33 | IT: 80.83
5. Below 60	`WHERE math<60 OR science<60 OR prog<60`	Rahul identified
6. Class Average	`AVG((math+science+prog)/3.0)`	77.53

Table 2 — SQL tasks with queries and results

3.2 Excel Analysis

- **Average Marks column** — Created using the formula =(D2+E2+F2)/3 applied across all rows.
- **Sorting** — Dataset sorted by Average Marks (Largest to Smallest) to rank students.
- **Conditional Formatting** — Marks below 60 highlighted in red across Math, Science, and Programming columns.
- **Pivot Table** — Department as Rows, Average Marks as Values (Average). Result: CS: 75.33, IT: 80.83, Grand Total: 77.53.

3.3 Python Analysis (Pandas)

The following Python code was used to load and analyze the dataset:

```
import pandas as pd

df = pd.read_csv("students.csv")
df["average_marks"] = (df["math"] + df["science"] + df["programming"]) / 3

top_student = df.loc[df["average_marks"].idxmax()]
dept_avg    = df.groupby("department")["average_marks"].mean()
correlation = df["attendance"].corr(df["average_marks"])

print("Top Student:", top_student["name"])
print("Dept Averages:\n", dept_avg)
print("Attendance-Marks Correlation:", round(correlation, 2))
```

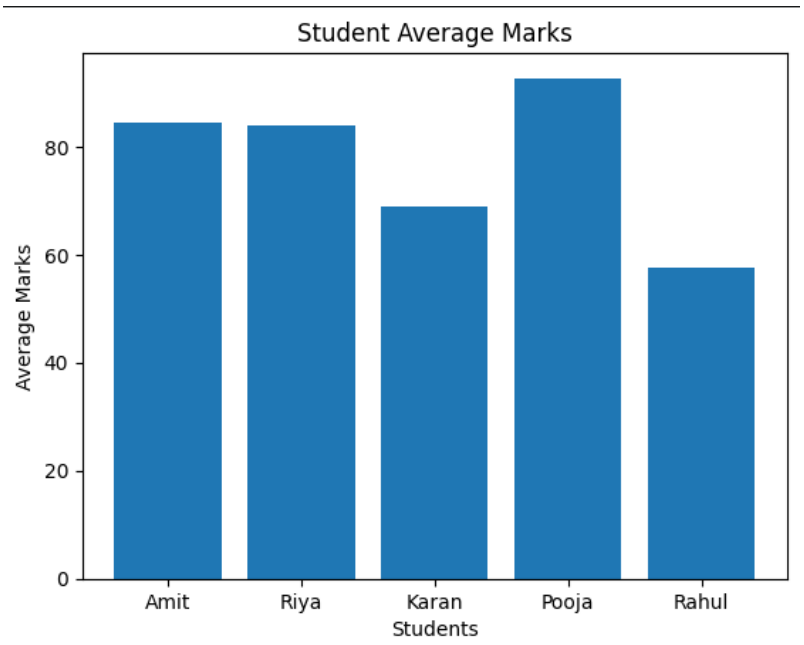
The script outputs the top-performing student (Pooja), department-wise averages, and a Pearson correlation coefficient between attendance and average marks.

4. Visualizations

Three charts were generated using Matplotlib to visually communicate the findings.

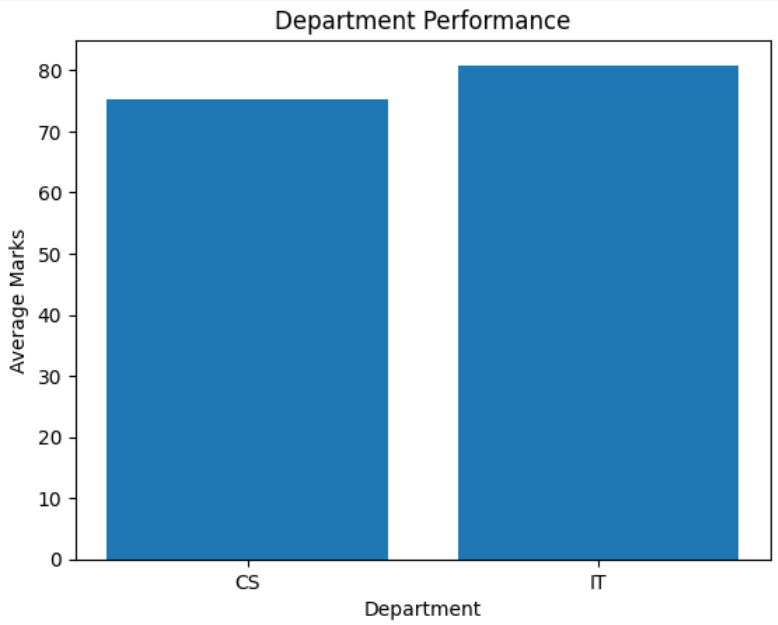
Bar Chart — Student Average Marks

A vertical bar chart with student names on the X-axis and average marks on the Y-axis. Pooja's bar is the tallest (92.67) and Rahul's is the shortest (57.67), making it easy to rank all students at a glance.



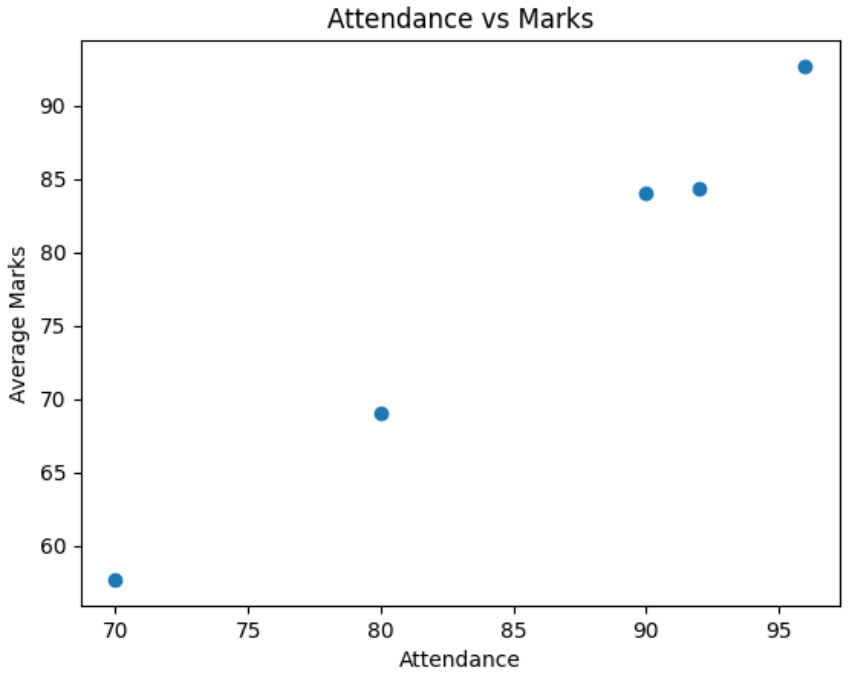
Bar Chart — Department Performance

A bar chart comparing CS and IT departments. IT (80.83) clearly outperforms CS (75.33). This chart helps the administration identify which department may need additional academic support.



Scatter Plot — Attendance vs Average Marks

Each point represents a student plotted by attendance (X-axis) against average marks (Y-axis). The upward trend confirms a positive correlation — students with higher attendance tend to score better.



5. Tech Stack

The following tools and technologies were used across all three phases of this project:

Tool / Technology	Purpose
Python 3	Core programming language for data analysis
Pandas	Data loading, manipulation, and grouping
Matplotlib	Chart generation — bar chart and scatter plot
Microsoft Excel	Spreadsheet analysis, pivot tables, conditional formatting
SQL (MySQL/SQLite)	Querying and aggregating student data
CSV	Dataset storage format (students.csv)

6. Key Findings

Department	Students	Avg Marks	Top Scorer	At-Risk
CS	Amit, Riya, Rahul (3)	75.33	Amit — 84.33	Rahul — 57.67
IT	Pooja, Karan (2)	80.83	Pooja — 92.67	None

Table 3 — Department-wise performance comparison

- **Pooja (IT)** is the top-performing student with an average of 92.67 marks and 96% attendance.

- **Rahul (CS)** is the only at-risk student, scoring below 60 in Mathematics (55) and Programming (58).
- **IT department** outperforms CS with an average of 80.83 vs 75.33.
- **Class average** is 77.53 marks across all five students.
- **Positive correlation** exists between attendance and marks — higher attendance consistently leads to better performance.
- All three tools (SQL, Excel, Python) produced consistent and matching results, validating the analysis.

7. Unique Points

- The same dataset was analyzed using three different tools — SQL, Excel, and Python — providing cross-validation of all results.
- Conditional formatting in Excel gives an instant at-a-glance visual for at-risk students without writing any code.
- The scatter plot reveals an attendance-performance relationship that is not obvious from the raw numbers alone.
- Python's `corr()` function quantifies the attendance-marks relationship as a Pearson correlation coefficient, adding statistical rigor.
- The Pivot Table in Excel is dynamic — more departments or students can be added and results update automatically.

8. Future Improvements

- **Larger Dataset** — Expand to include more students, subjects, and semesters for broader analysis.
- **Predictive Modelling** — Use machine learning (e.g., Linear Regression) to predict final grades from attendance and mid-term scores.
- **Interactive Dashboard** — Build a web-based dashboard using Plotly or Power BI for real-time filtering by department or subject.
- **Automated Alerts** — Set up automated email alerts when a student's marks drop below a defined threshold.
- **Database Integration** — Store data in a relational database (MySQL) for multi-year tracking and complex SQL queries.
- **Semester-wise Trends** — Add a time dimension to track whether student performance is improving or declining over semesters.